

ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY
MATHEMATICS DEPARTMENT
 Mathematics for Natural Science(math1011)(Test I(15%))

Part I(Short answer) Write your simplified answer on the space provided(1pts for each blank space)

1. The negation of $(\forall x)(3x + 1 < 0)$ is _____
2. All real number x and some real number y such that $x^2 - y^2 > 0$. The open statement symbolically is _____ and the truth value is _____.
3. Let $A = \{2x : -3 < x \leq 3, x \in \mathbb{Z}\}$ and $B = \{x : 0 < x < 6, x \in \mathbb{Z}\}$, then
 - a) List $p(A \cap B) =$ _____
 - b) $n(A/B) =$ _____
4. Let $\text{LCM}(a, 45) = 180$ and $\text{GCF}(a, 45) = 3$, then the value of $a =$ _____.
5. Let $z = \frac{2}{\sqrt{3}} - 2i$ and $w = -\frac{2}{\sqrt{3}} + 2i$, then
 - a) The principal argument of zw is _____
 - b) The polar form of z^4 is _____
 - c) Solve $w^{1/2}$. Ans: _____

Part II(Workout): Show all the necessary steps clearly and neatly. (2+2+2 pts)

1. Identify the premises, write the argument form and check the validity of the argument:
 If he studies medicine, he will get a good job. If he gets a good job, he will get a good wage. He did not get a good wage. Therefore, he did not study medicine.



AASTU
UNIVERSITY FOR INDUSTRY

COLLEGE OF NATURAL AND APPLIED SCIENCES

DEPARTMENT OF MATHEMATICS, PHYSICS AND STATISTICS

Test I (15%)

Time allowed: 50 minutes.

Mathematics for Natural sciences (Math 1007)

Date: January 17, 2024

Name: Yaphet Culilat

Id No.: 1408/1b

Sec: 22

Part I: Short answer questions. (1 pts.)

1. If the truth value of $(p \vee \neg q)$ is False, then the truth value of $\neg(p \wedge r) \Rightarrow (q \Rightarrow p)$ is True.
2. The truth value of $(\exists x)(\forall y)(x, y \in \mathbb{Z})(x - y = 2)$ is True.
3. If $s_n = \left\{4 - \frac{1}{2^n}\right\}_{n=0}^{\infty}$, then lub of $s_n = \underline{4}$ and glb of $s_n = \underline{3}$.
4. For any open proposition $p(x, y)$, $(\forall x)(\exists y)p(x, y) \Leftrightarrow (\exists x)(\forall y)p(x, y)$ is a tautology
(True/False) False.
5. The fractional representation of $87.124\overline{3264}$ is $\frac{871243264}{9999}$.
6. The square roots of the complex number $Z = -1 - i\sqrt{3}$ are $1 - i$ and $-1 + i$.
7. The most simplest form of $(A \cap B') \cup (A \cup B)'$ is A' .